AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 1 · STUDENT LESSON

A question that evidence can answer

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Name the cases. Define the outcome. Bound the question.

Turn broad aspirations into measurable comparisons while distinguishing practice snapshots from collected research evidence.

Keep one habit: Question protocol → constructed retrieval ledger → six-task diagram design → scope probes → scoring audit → independent speed/quality question.

MAIN LESSON · SECTION 1

Turn an aspiration into a question with a measurable answer

> Name the versions, cases, outcome rule and limits before comparing.

Learning goals

I can explain the **mechanism** and terminology of a question that **evidence** can answer.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve engineering evidence and bounded claims

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Engineering demonstration and defense. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall AI 6 exit **evidence**, reproducible experiments, probability and descriptive statistics. Statistical inference is taught with its assumptions before use.

  Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain a question that **evidence** can answer using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of a question that **evidence** can answer

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define the outcome and conditions that would answer the question

> A clear comparison can be investigated without promising the preferred result.

Begin with “useful for which task?”

“Prove AI is useful” sounds like a goal, but useful for whom and for which task? A research question needs an observable outcome and a fair comparison under stated conditions. For room retrieval, ask whether method B returns more relevant records than baseline A within the top three on the same five fixed queries. Define relevance and the eligible collection first.

That result could answer this local question without answering whether every AI system is useful.

Define an answerable research question

A **research question** identifies an **outcome**, comparison and conditions that **evidence** could answer. A broad aspiration needs narrowing before a study.

Complete fixed-query practice protocol

Question element	Complete added fictional practice protocol
Versions and comparison	Baseline A and method B; their ranking outputs are supplied snapshots, not implemented algorithms

Question element	Complete added fictional practice protocol
Cases	Fixed matched query IDs Q1–Q5, same for both versions
Eligible corpus	c1 contains R1–R10 only
Outcome	Number of supplied relevant records in the first three, averaged over five queries
Scoring rule	Fixed supplied relevance sets; exactly three unique eligible records per query
Relevance sets	Q1 R1/R2; Q2 R3; Q3 R4/R5; Q4 R6; Q5 R7/R8
What is held fixed	Case IDs, corpus/version, cutoff 3 and relevance labels
Evidence status	Added constructed snapshots for worked practice; no live retrieval or real annotator judgments
Reserved evaluation	A future unseen judged set must remain separate; these revealed examples are not fresh reserved cases
Claim boundary	Compare supplied outputs on these queries only; no universal usefulness, deployment impact, causal mechanism or child-learning claim

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured **outcome**.

MAIN LESSON · SECTION 3

Trace the question through a fully supplied practice snapshot

> B's higher mean here does not erase its worse Q4 result.

Original retrieval question and predeclared rules

For fictional classroom retrieval, ask whether method B returns more relevant records than lexical baseline A on a reserved judged query set at cutoff 3. Define eligible corpus and relevance rules before running.

Added constructed rankings, labels and complete counts

Query	Supplied relevant IDs	A top three / relevant count	B top three / relevant count
Q1	R1, R2	R1, R9, R10 / 1	R1, R2, R9 / 2
Q2	R3	R3, R9, R10 / 1	R3, R9, R10 / 1
Q3	R4, R5	R4, R9, R10 / 1	R4, R5, R9 / 2
Q4	R6	R6, R9, R10 / 1	R9, R10, R8 / 0
Q5	R7, R8	R7, R9, R10 / 1	R7, R8, R10 / 2
Five-query mean	Same fixed labels and corpus	5/5 = 1 relevant record	7/5 = 1.4 relevant records

State measurement assumptions before interpreting

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

A question that evidence can answer — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Proposal says prove AI is useful; available records are retrieval results for five room queries.	For fictional classroom retrieval, ask whether method B returns more relevant records than lexical baseline A on a reserved judged query set at cutoff 3. Define eligible corpus and relevance rules before running.	Narrow to relevant retrieval at a declared cutoff on those queries. The supplied records cannot answer universal usefulness or deployment impact.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict easier tasks given only to version B

> Changing task difficulty with version spoils the intended comparison.

Current task and response

- Proposal says prove AI is useful; available records are retrieval results for five room queries. Show the intermediate steps and give the conclusion with its **scope**.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the different-task change

- Change: Replace the six tasks with six easier tasks only for B.

▼ **Check after predicting**

Prediction: The comparison no longer isolates the diagram condition.

Reason: Task difficulty now changes alongside the proposed intervention.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the diagram question as a different six-task study design

> Distinguish a measurable question from evidence that has actually been collected.

Main broad-aspiration problem and bounded repair

Proposal says prove AI is useful; available records are retrieval results for five room queries.

Narrow to relevant retrieval at a declared cutoff on those queries. The supplied records cannot answer universal usefulness or deployment impact.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate diagram questions and complete comparison scope

A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

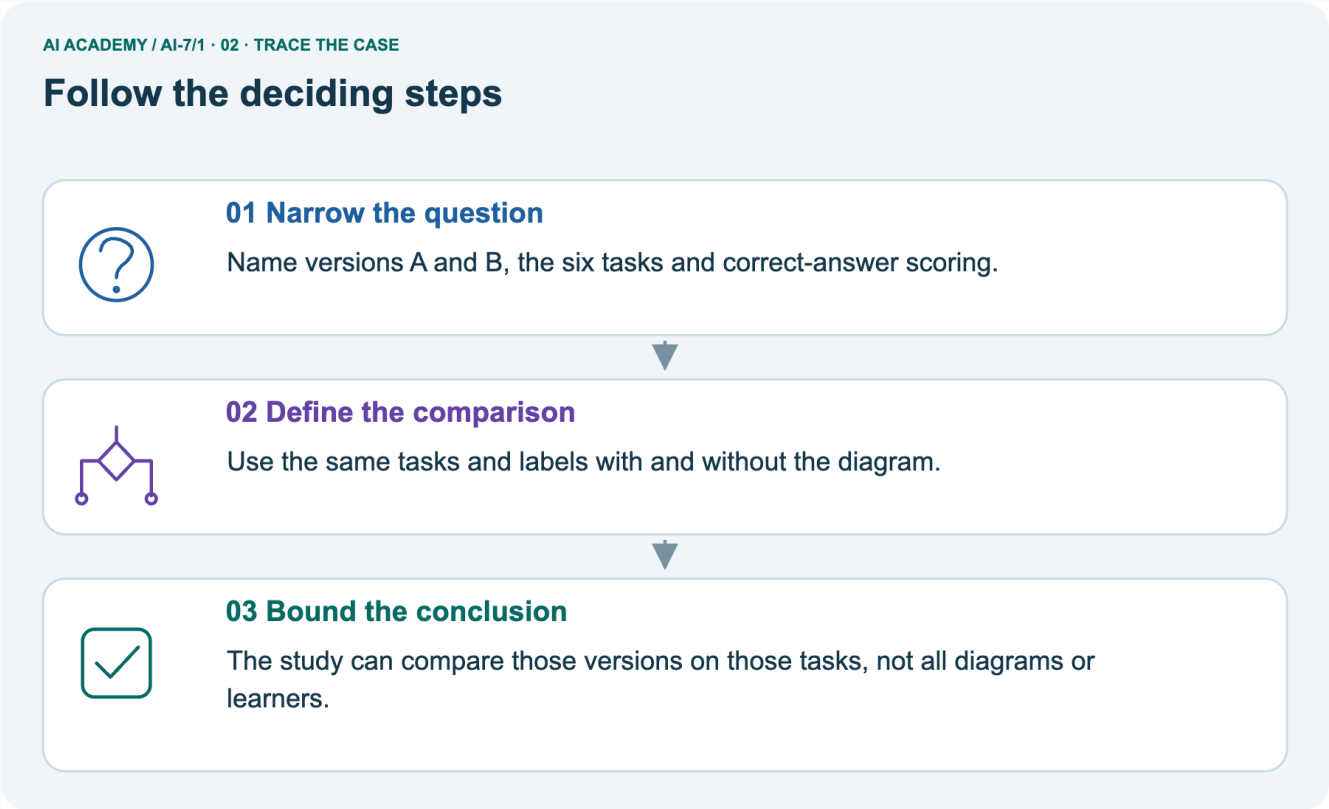
Candidate questions: "Are diagrams wonderful?" and "Does version B answer more of these six tasks correctly than version A?"

The picture asks a separate diagram question: version A without the diagram versus B with it on the same six fixed plant-identification tasks and answer labels. “Are diagrams wonderful?” has no defined measurement here. “Does B answer more of these six correctly than A?”

specifies versions, outcome and denominator. No actual answers or counts are supplied for the picture, so do not claim B wins.

If only B receives easier tasks, version and difficulty change together; the resulting difference cannot isolate the diagram comparison. Even a matched six-task result would not establish all diagrams' value or effects on children's learning.

Trace versions, fixed scoring and bounded conclusion



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Narrow the question	Name versions A and B, the six tasks and correct-answer scoring.
2. Define the comparison	Use the same tasks and labels with and without the diagram.
3. Bound the conclusion	The study can compare those versions on those tasks, not all diagrams or learners.

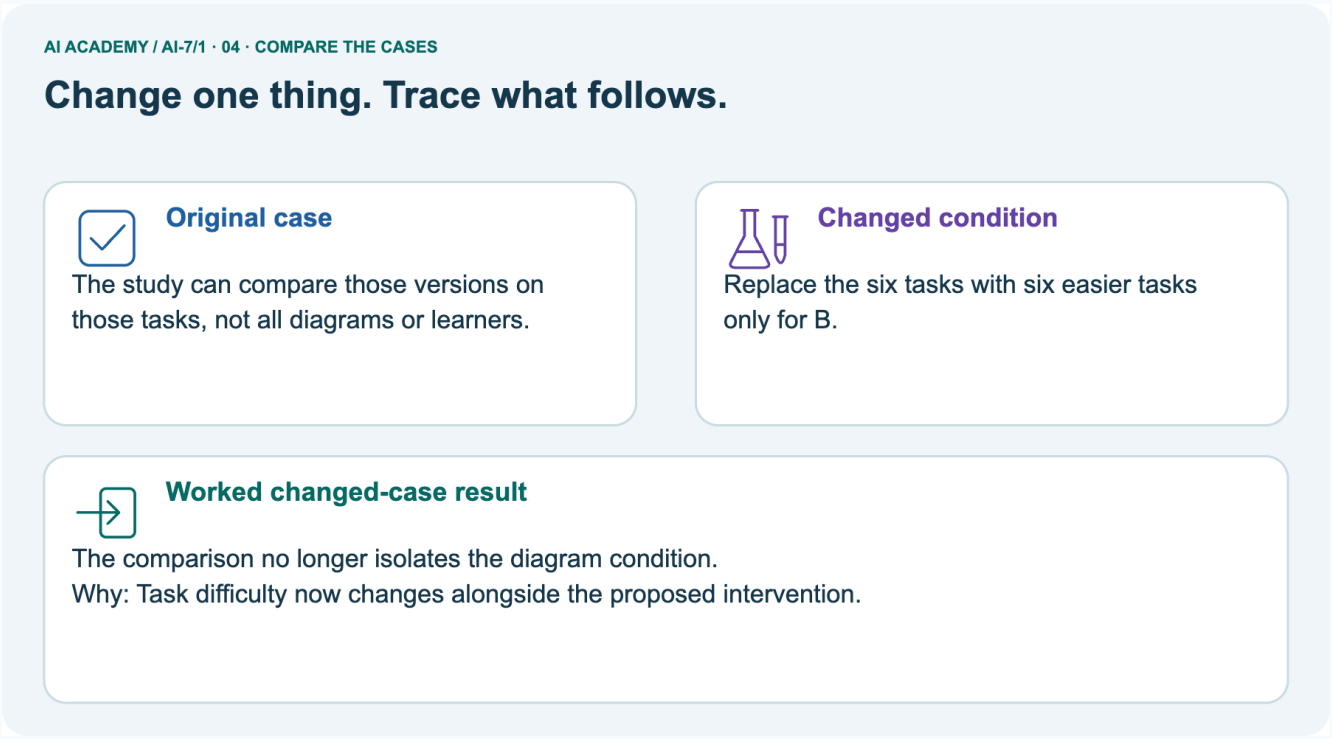
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An answerable **research question** connects a change to an observable **outcome** and a specified population. Opinion words need an operational definition before measurement.

Predict easier B-only tasks

- Change one thing: Replace the six tasks with six easier tasks only for B. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain missing results and child-learning limits

- This fictional study does not establish effects on children or general classroom learning.
- Read the labelled picture: Which proposed diagram question has a measurable **outcome**? Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.

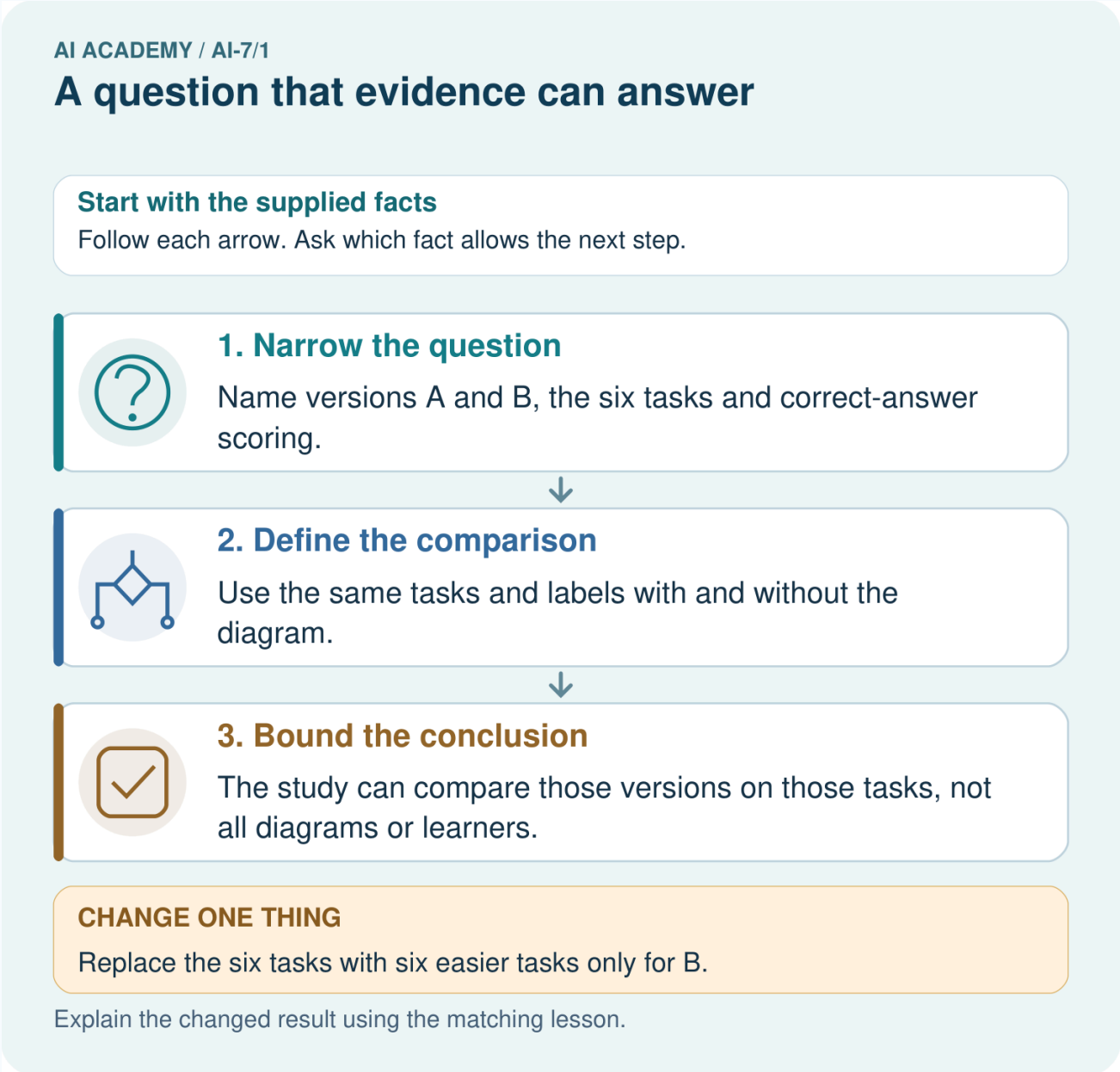
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: A question that evidence can answer

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit what the question and available evidence can support

> Specify missing details rather than filling them with an assumed result.

Core question-scope qualification and failure analysis

A measurable narrow question is not **evidence** of its answer.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete measurement and comparison probes

Research-question probe	Expected interpretation
“Is AI useful?”	Needs task, comparison, measurable outcome and conditions
A/B rankings on same five queries at cutoff 3	Can compare relevant counts under supplied labels
B wins mean but loses Q4	Report both; mean does not mean every case improved
B only receives easier tasks	Comparison confounds version with task difficulty
Rubric/corpus changes between versions	Different measurement conditions require review
No outcome records yet	Question can be designed; no empirical winner can be stated
Practice examples already revealed	Do not call them unseen reserved evaluation
Five or six fictional cases	No population-wide usefulness or real learner effect follows

Research-question probe	Expected interpretation
A better score is observed	Does not by itself identify which internal mechanism caused the change

Guided repair and explanation

- A report omits this qualification: A measurable narrow question is not **evidence** of its answer. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise naming the comparison, outcome and fixed conditions

> Write a question that evidence could answer even if B does not win.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example “A question that **evidence** can answer”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for A question that **evidence** can answer. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Which proposed diagram question has a measurable **outcome**? Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the selected ten-request latency/correctness case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run the complete fixed-snapshot scoring investigation

> Use computation to check the observable outcome without inventing a broader study.

Current broad-proposal investigation

Proposal says prove AI is useful; available records are retrieval results for five room queries. Show the intermediate steps and give the conclusion with its **scope**.

Runnable constructed rankings and relevance audit

```
"""Added constructed retrieval snapshot for a bounded research question, no live
retrieval."""
from fractions import Fraction
CORPUS={'R'+str(i) for i in range(1,11)}
# Relevance labels are a supplied classroom rubric, not real annotator judgments.
RELEVANT={'Q1':{'R1','R2'},'Q2':{'R3'},'Q3':{'R4','R5'},'Q4':{'R6'},'Q5':{'R7','R8'}}
A={'Q1':['R1','R9','R10'],'Q2':['R3','R9','R10'],'Q3':['R4','R9','R10'],
   'Q4':['R6','R9','R10'],'Q5':['R7','R9','R10']}
B={'Q1':['R1','R2','R9'],'Q2':['R3','R9','R10'],'Q3':['R4','R5','R9'],
   'Q4':['R9','R10','R8'],'Q5':['R7','R8','R10']}

def score_snapshot(ranking,labels,corpus,cutoff=3):
 if type(cutoff) is not int or cutoff!=3:raise ValueError('this classroom protocol fixes
cutoff at exactly three')
 if not ranking or set(ranking)!=set(labels):raise ValueError('matched nonempty query IDs
and supplied labels required')
 counts={}
 for query,ids in ranking.items():
  if type(ids) is not list or len(ids)!=cutoff or len(set(ids))!=cutoff or not set(ids)
<=corpus:
   raise ValueError('exact unique eligible top-k records required')
  if type(labels[query]) is not set or not labels[query]<=corpus:raise
ValueError('eligible relevance set required')
  counts[query]=sum(rid in labels[query] for rid in ids)
 return
{'counts':counts,'mean_relevant_in_top3':Fraction(sum(counts.values()),len(counts)),
         'queries':len(counts),'cutoff':cutoff}
a=score_snapshot(A,RELEVANT,CORPUS);b=score_snapshot(B,RELEVANT,CORPUS)
assert list(a['counts'].values())==[1,1,1,1,1] and list(b['counts'].values())==[2,1,2,0,2]
```

```
assert a['mean_relevant_in_top3']==1 and b['mean_relevant_in_top3']==Fraction(7,5)
assert b['counts']['Q4']<a['counts']['Q4']
print('Bounded question: On these five fixed queries, with corpus c1 and supplied
relevance rules, does B return more relevant records in the top three on average than A?')
print('Added constructed snapshot A:',a,'B:',b)
print('B has larger mean here, but worsens Q4; no universal usefulness, causal mechanism,
deployment impact or learner outcome follows.')
print('All rankings and labels are supplied fictional inputs; no retriever, model, child,
human annotator or experiment was run. These revealed cases are practice, not a fresh
reserved evaluation.')
```

Expected counts and evidence status

1. Write the bounded question and predict relevant counts from the supplied table before running or tracing. A has five counts of 1; B has 2, 1, 2, 0, 2, so the means are 1 and 1.4. Keep Q4's decline visible. The code scores supplied rankings against fixed labels and rejects mismatched IDs, duplicate/ineligible records and a cutoff other than three.

   No algorithm actually retrieves or generates an answer, and no person supplies new judgments. These fully revealed constructed cases are practice, not a fresh experiment or reserved evaluation. The picture's six-task diagram study has no result records and cannot borrow these retrieval numbers.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a ten-request speed comparison with correctness preserved

> Name the timing/scoring details that still need to be fixed.

Complete selected independent task

You have response times and correctness for two lookup procedures on ten fictional requests. Write a bounded question and one missing detail.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

You have response times and correctness for two lookup procedures on ten fictional requests. Write a bounded question and one missing detail.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the question, measurement rule and intended evidence

> Separate planned study, constructed practice and actual observations.

Project task and artifact

Add a labelled a question that **evidence** can answer **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of a question that **evidence** can answer.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual scoring run or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the bounded question and one missing piece of evidence

> A plan can be answerable before any result exists.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of a question that **evidence** can answer

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Research question	A bounded question tied to observable evidence
Outcome	The quantity or behaviour being measured
Scope	The conditions covered by a question

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A measurable narrow question is not **evidence** of its answer. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected **outcome** before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the **outcome** from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected **outcome** before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

A question that evidence can answer — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Proposal says prove AI is useful; available records are retrieval results for five room queries.	For fictional classroom retrieval, ask whether method B returns more relevant records than lexical baseline A on a reserved judged query set at cutoff 3. Define eligible corpus and relevance rules before running.	Narrow to relevant retrieval at a declared cutoff on those queries. The supplied records cannot answer universal usefulness or deployment impact.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

By the end, I can	How I will show it
Explain the mechanism and terminology of a question that evidence can answer	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.
- Which labelled step matches this detail? The study can compare those versions on those tasks, not all diagrams or learners.
- 1. Define the comparison
- 2. Bound the conclusion
- 3. Narrow the question
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Bound the conclusion The study can compare those versions on those tasks, not all diagrams or learners.

An answerable **research question** connects a change to an observable **outcome** and a specified population. Opinion words need an operational definition before measurement.

Changed-case question and optional hint

- Changed-case practice: Replace the six tasks with six easier tasks only for B. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The comparison no longer isolates the diagram condition.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The comparison no longer isolates the diagram condition. Task difficulty now changes alongside the proposed intervention.

Repair and retain the record

Review Engineering demonstration and defense (ai-6/36). Goal: Explain the **mechanism** and terminology of engineering demonstration and defense. Short practice: Locate V7, room E and read-only **scope** in the demonstration.

Which **evidence** supports the answer, and why must a request to email everyone still be refused?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work
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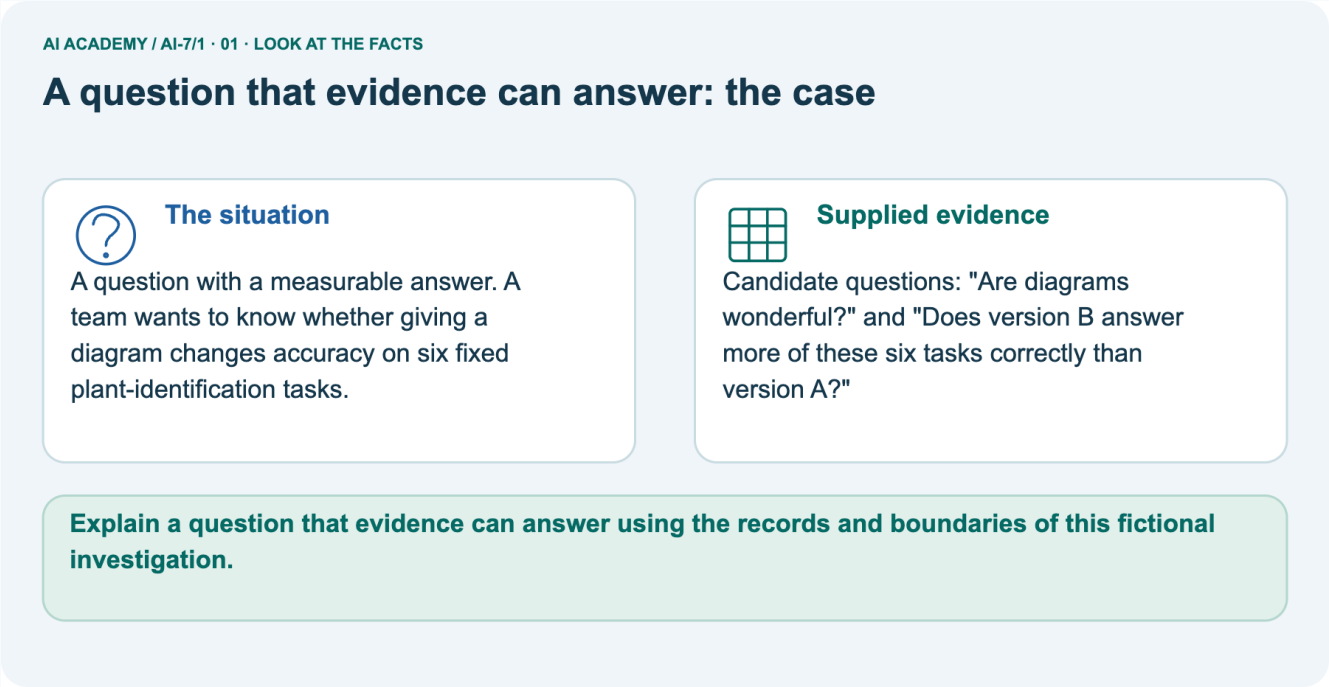
<details>
<summary>Reference · Six-picture concept journey</summary>
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REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. A question that evidence can answer: the case



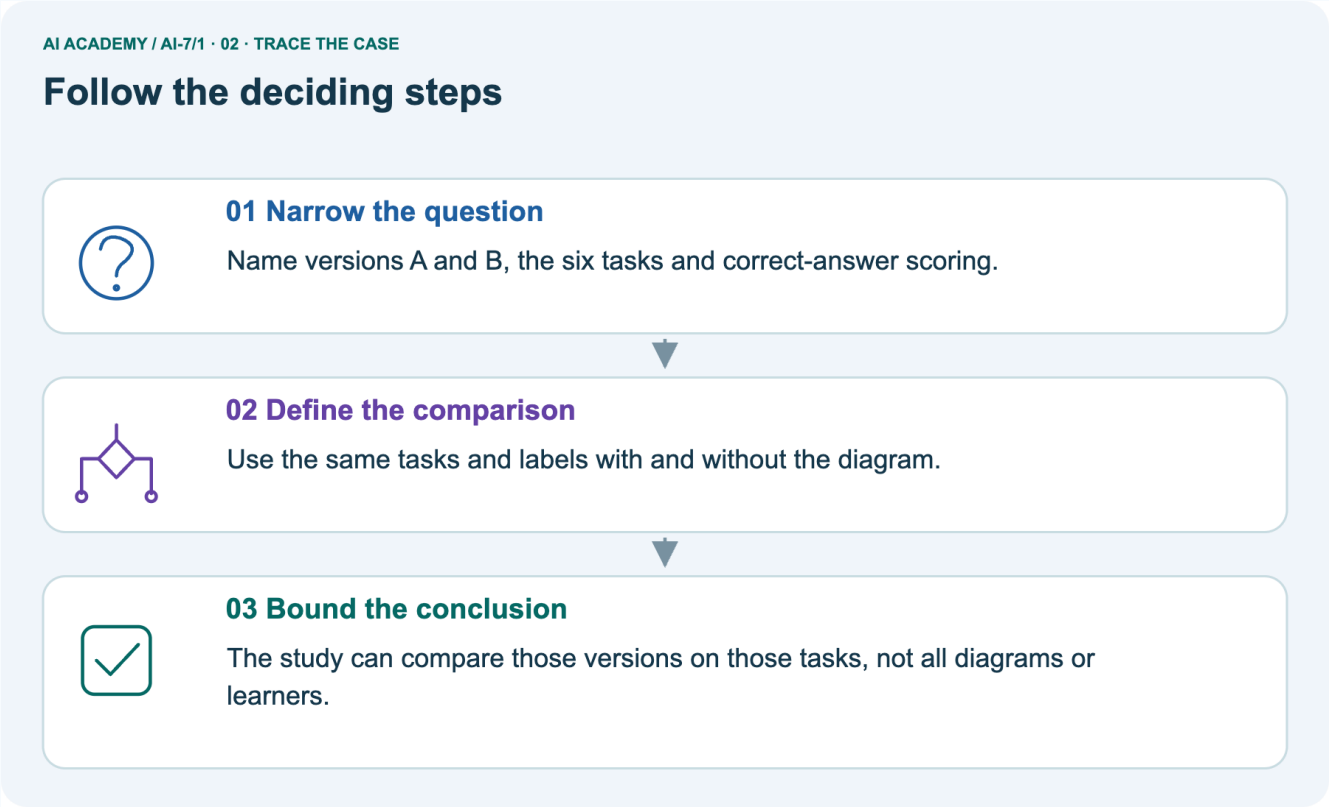
Follow the labelled picture with the matching explanation. Open the vector diagram

A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

Supplied facts: Candidate questions: "Are diagrams wonderful?" and "Does version B answer more of these six tasks correctly than version A?"

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

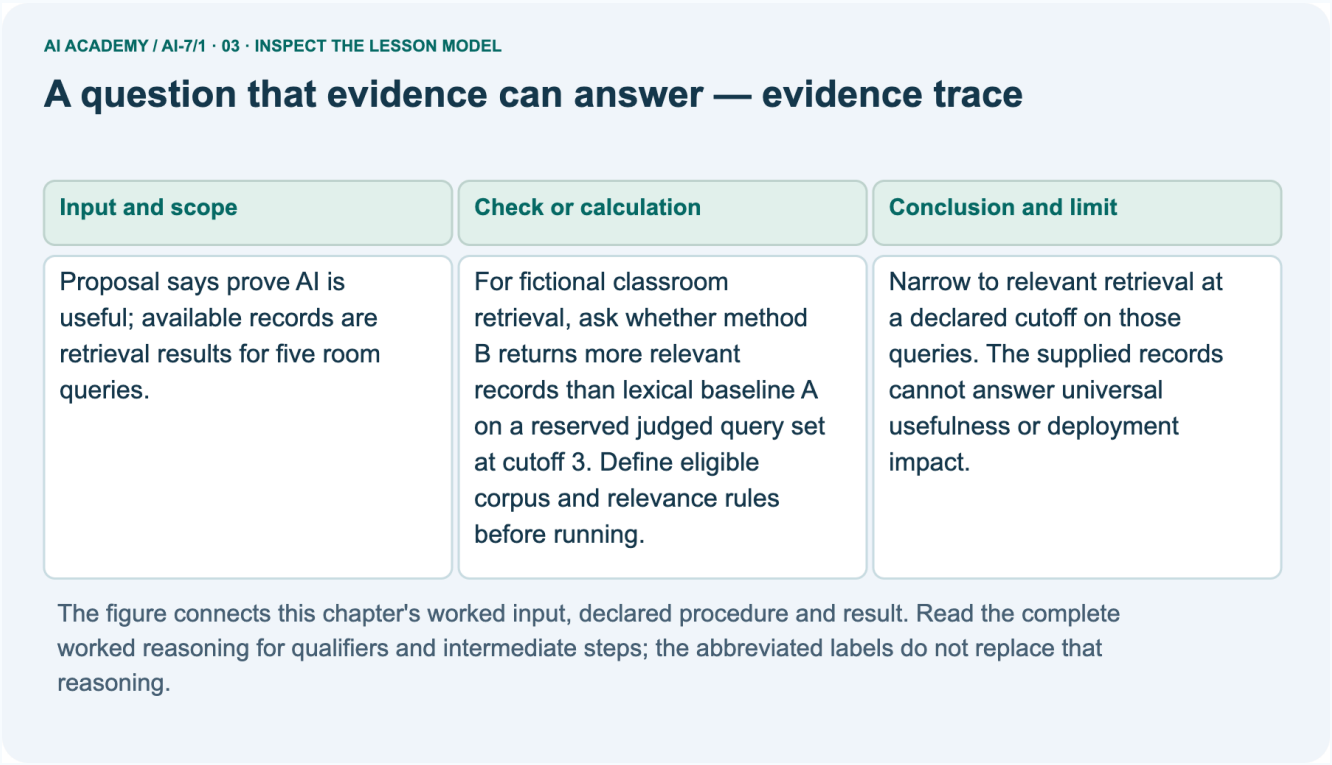


Follow the labelled picture with the matching explanation. Open the vector diagram

Narrow the question: Name versions A and B, the six tasks and correct-answer scoring. → Define the comparison: Use the same tasks and labels with and without the diagram. → Bound the conclusion: The study can compare those versions on those tasks, not all diagrams or learners.

Point to each numbered stage and explain which recorded **input** it uses.

03. A question that evidence can answer — evidence trace

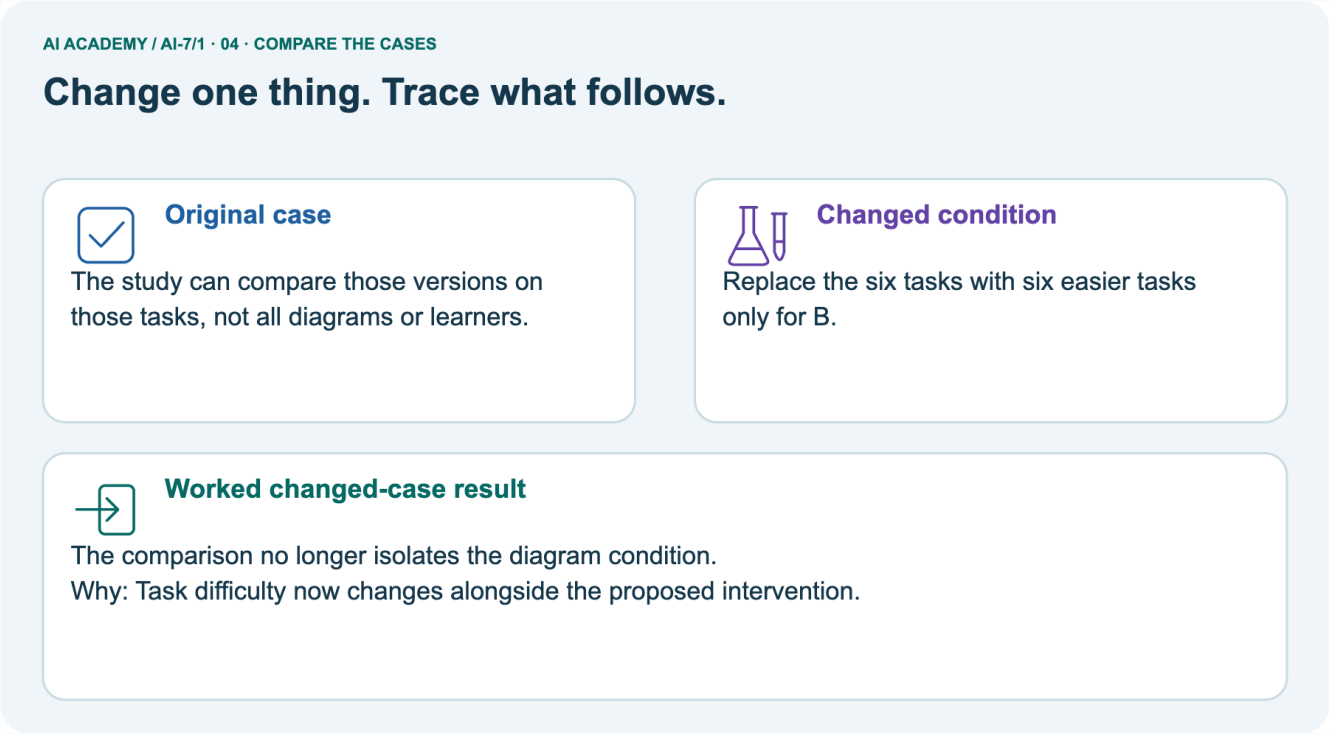


Follow the labelled picture with the matching explanation. Open the vector diagram

A question that **evidence** can answer — **evidence** trace. Proposal says prove AI is useful; available records are retrieval results for five room queries. | For fictional classroom retrieval, ask whether method B returns more relevant records than lexical baseline A on a reserved judged query set at cutoff 3. Define eligible corpus and relevance rules before running. | Narrow to relevant retrieval at a declared cutoff on those queries. The supplied records cannot answer universal usefulness or deployment impact.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



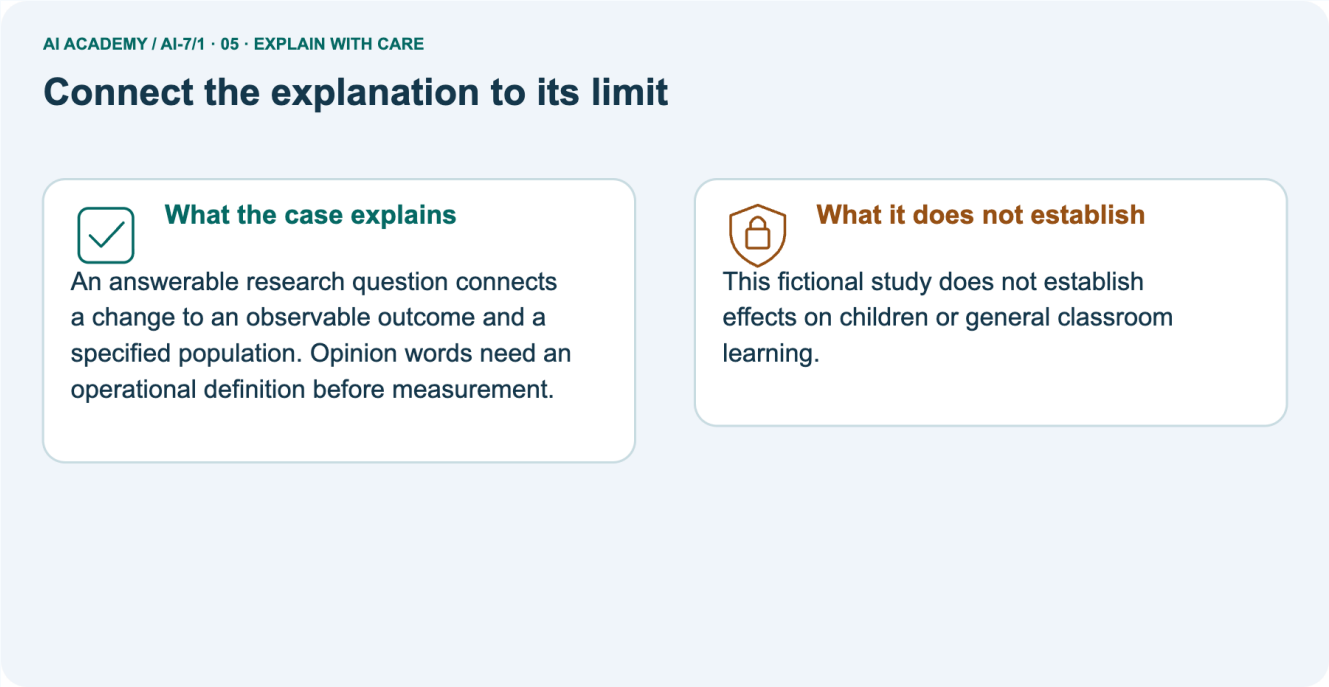
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The study can compare those versions on those tasks, not all diagrams or learners. Changed condition: Replace the six tasks with six easier tasks only for B.

Expected result: The comparison no longer isolates the diagram condition. Reason: Task difficulty now changes alongside the proposed intervention.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



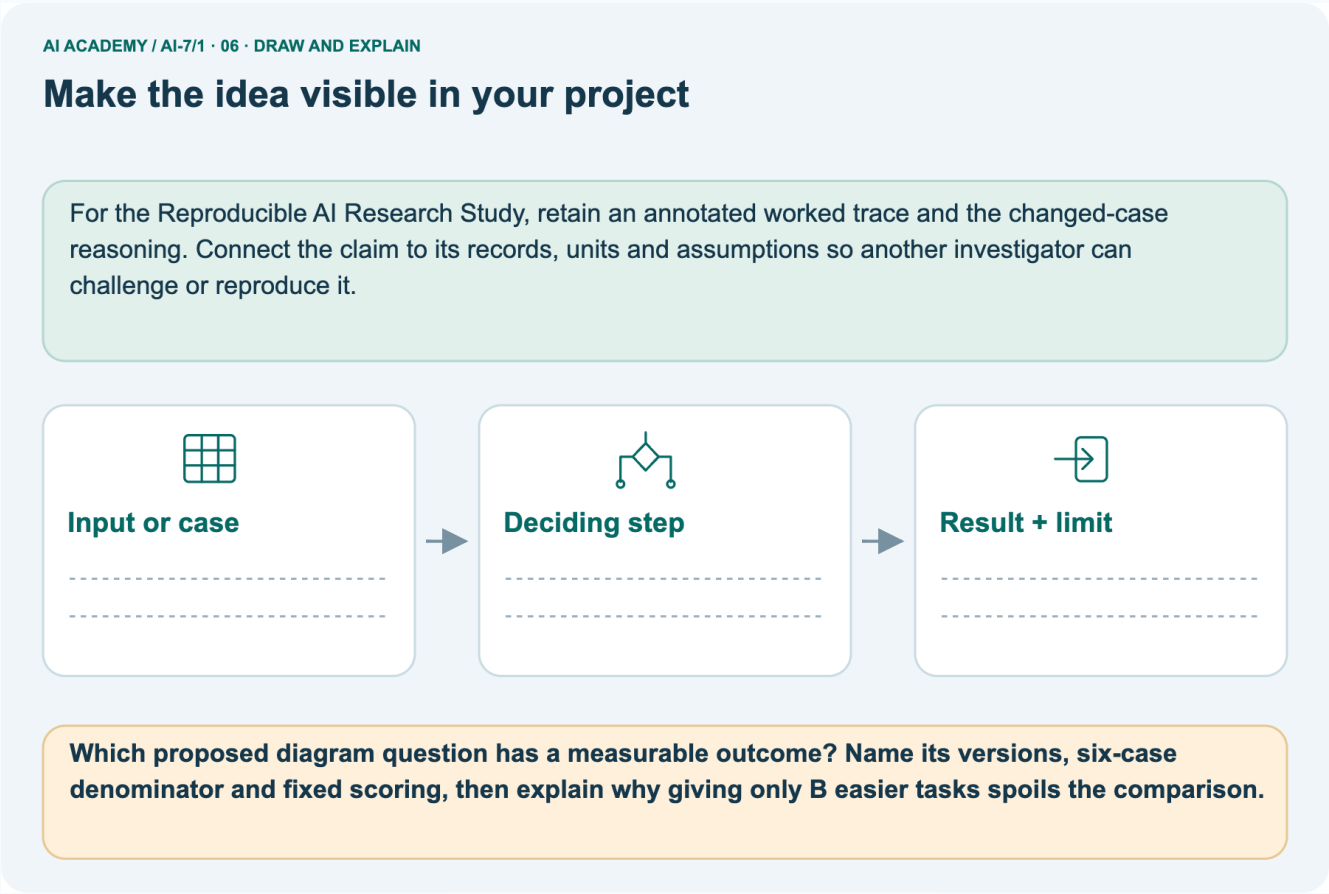
Follow the labelled picture with the matching explanation. Open the vector diagram

An answerable **research question** connects a change to an observable **outcome** and a specified population. Opinion words need an operational definition before measurement.

Limit: This fictional study does not establish effects on children or general classroom learning.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Which proposed diagram question has a measurable **outcome**?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. A question that evidence can answer: the case

02. Follow the deciding steps

03. A question that evidence can answer — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.